

Re-evaluating the Sun Liu Alliance in the Battle of Red Cliffs

A Spatial and Network Analysis of Lu Su's Brokerage Role

Digital Humanities Research Report

Primary Corpus: *Romance of the Three Kingdoms*, Chapters 43 through 50

Methods: Network Analysis (R / tidygraph / ggraph) & Spatial GIS Analysis (QGIS)

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Abstract

This paper presents a hybrid computational humanities investigation into the Sun Liu alliance that shaped the outcome of the Battle of Red Cliffs, as narrated in Chapters 43 through 50 of Luo Guanzhong's *Romance of the Three Kingdoms*. Combining force-directed network analysis with GIS-based spatial analysis, we construct two interlocking models: a person to person interaction network (37 character nodes) and a person to place spatial interaction network (31 location nodes). Contrary to narrative prominence, which foregrounds Zhuge Liang and Zhou Yu as the alliance's dominant figures, eigenvector centrality analysis reveals Lu Su as the true structural broker of the coalition. Spatial analysis corroborates this finding: Lu Su's movement trajectories constitute the primary logistical thread connecting Wu and Shu territorial spheres. We argue that the Sun Liu alliance should be reread not as a fusion of strategic will, but as a *fragile convergence* held together by one diplomat whose labour is rendered invisible by the narrative's emphasis on martial spectacle. The study demonstrates the capacity of Digital Humanities methods to surface latent structural patterns that close reading alone cannot recover.

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1. Introduction

The Battle of Red Cliffs (208 CE) is one of the most celebrated military confrontations in Chinese history and historical fiction. In Luo Guanzhong’s *Romance of the Three Kingdoms* (*Sanguozhi Yanyi*), Chapters 43 through 50 dramatise the formation and execution of the Sun Liu alliance against Cao Cao’s vastly superior northern armies. Scholarship on this episode has often privileged its two most theatrically visible protagonists: the Shu strategist Zhuge Liang and the Wu commander Zhou Yu, while treating the diplomat Lu Su as a secondary facilitator. This paper suggests that such a reading reflects the narrative emphasis of the novel. We believe that network and spatial methods can help us notice a different pattern of relationships and influence from traditional perspectives.

Thus, our research question is: *How do spatial map analysis and diplomatic network analysis reveal the hidden labour and fragile trust required to sustain the Sun Liu alliance in the Battle of Red Cliffs (Chapters 43 through 50)?* We hypothesise that this alliance is best understood not as a stable merger of factions but as a *fragile convergence*: a temporary alignment sustained by diplomatic labour that the narrative itself tends to obscure.

To explore this, we use (1) LLM-assisted character-interaction extraction, (2) force-directed network analysis and (3) GIS-based spatial analysis. Taken together, these methods point toward a more nuanced understanding of Lu Su’s role in the alliance.

The paper proceeds as follows. Section 2 surveys relevant scholarship in Digital Humanities, computational literary analysis and spatial humanities. Section 3 outlines our corpus, data extraction process, network construction and GIS procedures. Section 4 presents the network and spatial findings. Section 5 interprets these results through the lens of brokerage and fragile alliance dynamics. Section 6 discusses the study’s methodological limitations. Section 7 concludes by reflecting on the wider implications of this approach for reading pre-modern Chinese narrative.

2. Literature Context

2.1. Digital Humanities and Distant Reading

The application of computational methods to literary corpora gained sustained theoretical justification in Franco Moretti’s programme of “distant reading,” which advocated graph-based, map-based and tree-based models as alternatives to the canonical close reading of individual texts (Moretti 2005). Moretti’s later essay “Network Theory, Plot Analysis” explicitly imported graph-theoretic vocabulary into literary criticism, treating character interactions as edges and figures as nodes (Moretti 2013). Although these case studies

focus on Western dramatic and novelistic traditions, the framework is useful for analysing Chinese classical fiction, where ensemble casts create especially dense network structures.

2.2. Network Analysis in Literary Studies

The operationalisation of character networks has proceeded rapidly since Moretti’s initial sketches. Painter, Daniels, and Jost (2019) provide a methodological survey of network analysis for the digital humanities, distinguishing co-occurrence, interaction, and influence networks and cataloguing metrics (degree, betweenness, and eigenvector centrality) productive for humanistic interpretation. Their overview of centrality measures underscores that local metrics such as degree must be read alongside broader structural indicators, a caution directly relevant to our findings about Lu Su versus Zhuge Liang.

Applied to Chinese fiction specifically, Fan and Li (2021) constructed co-occurrence networks for the full text of *Romance of the Three Kingdoms*, computing network features across more than 1,000 character nodes. Their cluster analysis confirmed that the novel’s social topology is highly modular along factional lines (Wei, Shu, Wu). Our study adopts a comparable network perspective but uses manually verified interaction edges for a single episode rather than automated co-occurrence across the full novel. By restricting our corpus to Chapters 43 through 50, we retain the episodic granularity needed to track alliance formation within a specific narrative sequence.

Zhang et al. (2021) compared character-relation networks in the *Records of the Three Kingdoms* and the *Romance*, using deep-learning NLP to extract 93% of characters from the former and 91% from the latter. Their finding is that the *Romance* network is “more complex and dynamic” than the historical *Records*. This suggests that the novel actively elaborates diplomatic relationships beyond what the historical record supports, a caution directly relevant to our analysis of Lu Su, whose importance is better captured by structural position than by raw frequency of interaction.

2.3. Spatial Humanities and GIS

The integration of geographic information systems into humanistic inquiry has moved from cartographic illustration toward analytic spatial modelling. I. N. Gregory and Geddes (2014) helped consolidate spatial humanities research, emphasising qualitative spatial experiences encoded in historical and literary sources rather than point-maps alone.

Murrieta-Flores, Donaldson, and I. Gregory (2017) argue that point-based cartographic representations are inadequate for modelling qualitative human phenomena, and demonstrate how cost-surface analysis and least-cost-path analysis can support “more nuanced interpretations of historical works of travel writing and topographical literature.” This informs our decision to treat GIS outputs as interpretive complements to, rather than

replacements for, close narrative reading, while following their precedent for route reconstruction of character movement across the Yangtze region.

Together, these strands of scholarship position our project at the intersection of computational literary analysis and spatial humanities, fields that remain underdeveloped in studies of pre-modern Chinese narrative.

3. Methodology

3.1. Corpus and Data Extraction

Our primary corpus comprises Chapters 43 through 50 of the Moss Roberts English translation of *Romance of the Three Kingdoms* (Luo 1994), which cover the Sun Liu diplomatic negotiations, the fire-attack strategy at Red Cliffs and its immediate aftermath.

Character interactions were extracted using a two-stage pipeline. In the first stage, we prompted a large language model (GPT-4o) to identify all dyadic interaction events in each chapter, recording the interacting pair, the interaction type from a controlled vocabulary of seven categories (Diplomatic, Strategic, Command, Persuasion, Conflict, Deception, Social), the sentiment valence (Cooperative = +1; Neutral = 0; Hostile = -1) and the location in which the interaction occurs. In the second stage, a research assistant manually verified all extracted interactions against the source text, correcting misattributions, resolving name-alias ambiguities (e.g., “Kongming” $\equiv$ “Zhuge Liang”), and flagging uncertain extractions.

The dataset contains **37 person nodes**, **31 spatial nodes** and **two edge types**:

- **Person to person edges** weighted by interaction type and sentiment.
- **Person to place edges** recording character presence and activity at each location

3.2. Network Analysis

The network was constructed and analysed in R using the tidygraph and ggraph packages. The person-person interaction network was designed to capture both the strength and tone of ties between characters. Edge weights and sentiment scores were defined first and these edge attributes were then used to support node-level centrality analysis.

Edge Weights. We assigned interaction-type weights to reflect political and narrative consequence:

- Strategic / Diplomatic / Command: weight = 2.0
- Persuasion / Conflict / Deception: weight = 1.0
- Social: weight = 0.5

Sentiment Score. Each edge carries a continuous sentiment score in $[-1, +1]$, the mean of all cooperative/neutral/hostile valence labels assigned to interactions along that edge.

Centrality Measures. For each character node, we computed two centrality measures: weighted degree centrality and eigenvector centrality.

Weighted Degree Centrality. Weighted degree centrality measures the total strength of a node’s direct interaction ties by summing the weights attached to its edges. In this study, it captures narrative visibility by showing how frequently and how strongly a character is connected to others in the interaction network.

Eigenvector Centrality. Eigenvector centrality is a recursive measure that gives higher scores to nodes connected to other highly connected nodes. It captures structural influence rather than just direct interaction count.

Visualisation. The final network was drawn using a stress layout to improve separation between dense clusters. Node size represents eigenvector centrality, node colour represents faction, edge width represents aggregate interaction weight and edge colour represents sentiment on a red-to-grey-to-green scale. Nodes with no recorded interactions were excluded from the final plot.

The resulting graph is shown in Figure 1.

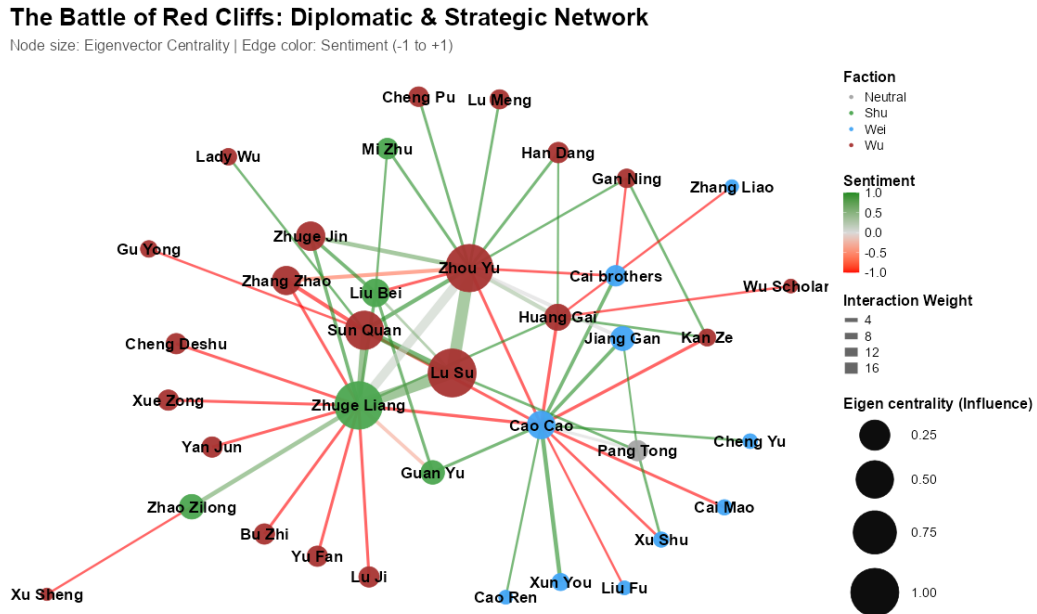


Figure 1: Person-person interaction network for **The Battle of Red Cliffs: Diplomatic and Strategic Network**. Node size represents eigenvector centrality, node colour represents faction, edge width represents interaction weight and edge colour represents sentiment

3.3. Spatial GIS Analysis

Geographic analysis was conducted in QGIS, projecting narrative locations onto a modern basemap using approximate coordinate assignments derived from cross-referencing the historical geography of the Eastern Han / Three Kingdoms period. Coordinate uncertainty was recorded as a categorical field (High / Medium / Low confidence) and propagated into spatial aggregations.

Three spatial outputs were produced:

1. **Interaction heat map:** each person to place edge contributes a weighted point event; kernel density estimation reveals the spatial distribution of coalition activity. See Figure 2
2. **Route reconstruction map:** character movement between successive locations is modelled as straight-line segments (geodesics), colour-coded by character, forming a trajectory overlay. See Figure 3.
3. **Timeline route visualisation:** movement segments are ordered chronologically by chapter section, allowing reading of spatial dynamics as narrative progresses. See Figure 4.

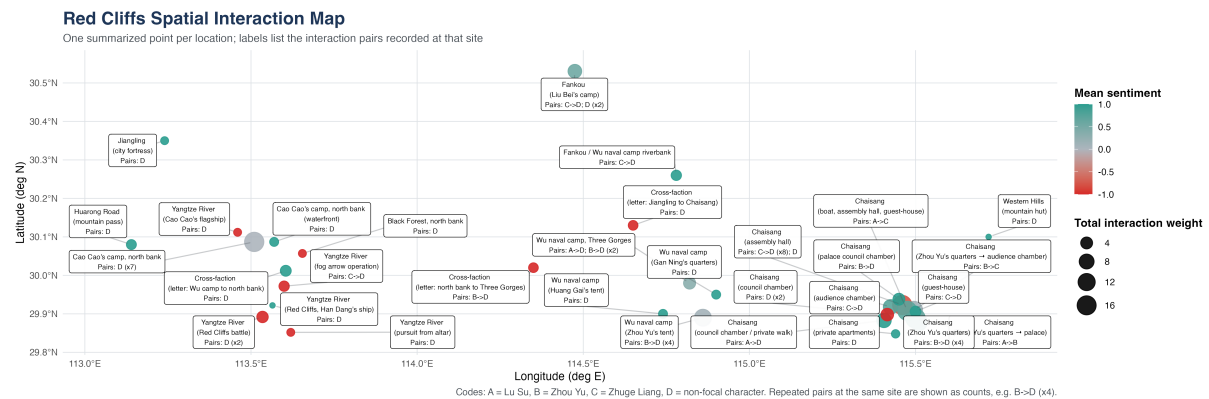


Figure 2: GIS-based spatial interaction map for Chapters 43 through 50. Warmer regions indicate higher narrative activity density. The Yangtze River functions as both a geographic and political boundary in the narrative.

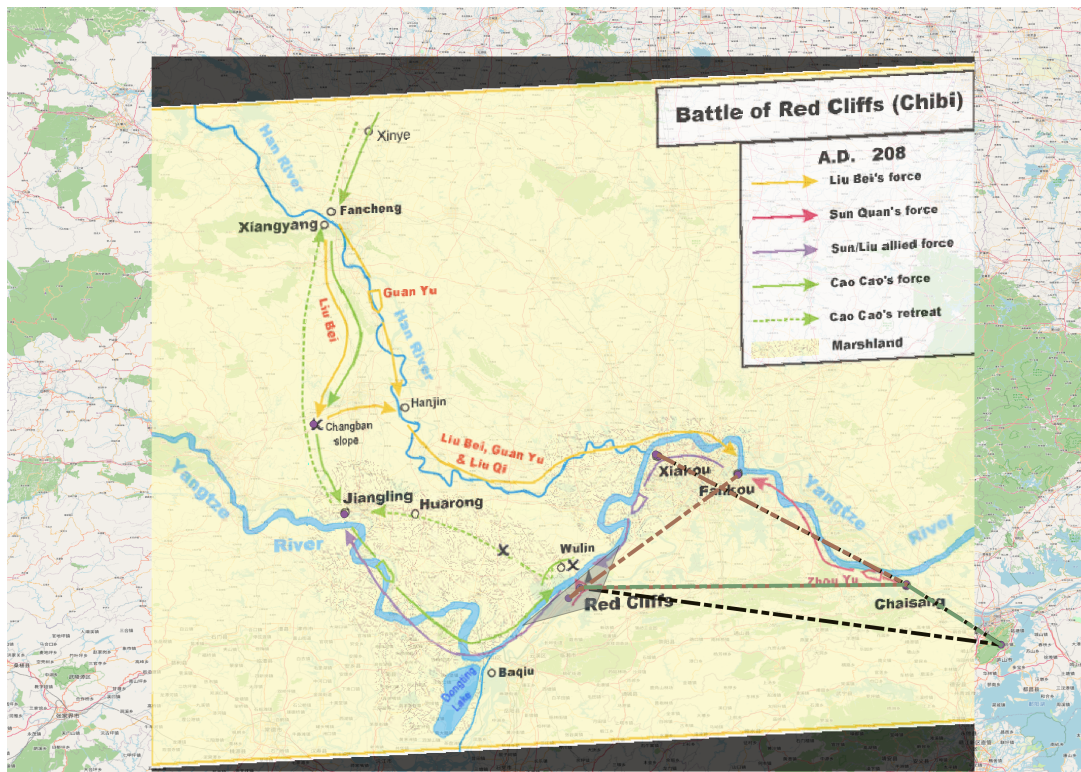


Figure 3: QGIS layer on modern basemap

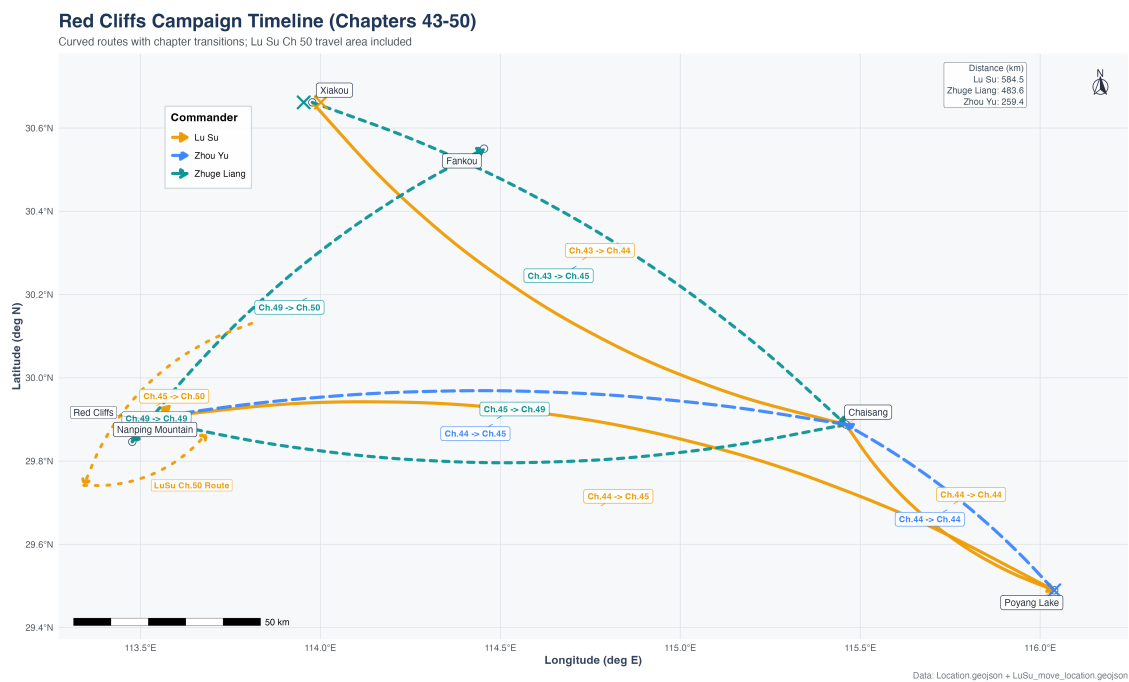


Figure 4: Timeline-based character movement route map. Lu Su's trajectory spans the greatest lateral distance, crossing the Wu and Shu spatial boundary multiple times. Zhou Yu remains anchored in Wu naval territory; Zhuge Liang exhibits long-range but episodic mobility.

3.4. R Code: Key Functions and Representative Snippets

The listings below are pseudocode-style summaries of the analysis pipeline. Full scripts (network plot, route reconstruction, timeline map) are available in the project repository.

```
1 # PSEUDOCODE: see scripts/network_analysis_red_cliffs.R
2 library(tidyverse)
3 library(tidygraph)
4
5 nodes <- read_csv("Nodes_table_Red_Cliffs.csv")
6 edges <- read_csv("Edges_table_Red_Cliffs.csv")
7
8 person_edges <- edges %>%
9   filter(Type == "Human interaction") %>%
10   rename(from = Nodes_A, to = Nodes_B)
11
12 rc_graph <- tbl_graph(nodes, person_edges, directed = FALSE)
13   %>%
14   activate(nodes) %>%
15   mutate(
16     degree_score = centrality_degree(weights = Weight),
17     eigen_score = centrality_eigen(weights = Weight))
```

Listing 1: Network construction and centrality (tidygraph); abbreviated pipeline.

The network figure (Figure 1) is produced with `ggraph()` and a stress layout: node size encodes eigenvector centrality, node colour faction, edge width interaction weight and edge colour mean sentiment.

```
1 # PSEUDOCODE: see scripts/map1_interaction_weighted_points.R
2 library(tidyverse)
3 library(sf)
4 library(ggplot2)
5
6 df <- read_csv("red_cliffs_with_locations.csv")
7 # assign proxy lon/lat per Location_Name (resolve_proxy_coord)
8
9 spatial_df <- df %>%
10   left_join(location_proxy, by = "Location_Name") %>%
11   group_by(Location_Name, lon, lat) %>%
12   summarise(total_weight = sum(Weight),
13             mean_sentiment = mean(Sentiment), .groups = "drop")
14
15 pts_sf <- st_as_sf(spatial_df, coords = c("lon", "lat"), crs =
```

```

4326)
16
17 ggplot() +
18   geom_sf(data = pts_sf, aes(size = total_weight, colour =
    mean_sentiment)) +
19   coord_sf(expand = FALSE)

```

Listing 2: Map 1: interaction map (sf / ggplot2); abbreviated pipeline.

```

1 # PSEUDOCODE: see scripts/map2_route_reconstruction.R
2 library(sf)
3
4 loc <- st_read("Location.geojson", quiet = TRUE)
5
6 linestring_from_places <- function(place_sf, ordered_names) {
7   m <- st_coordinates(place_sf[match(ordered_names,
    place_sf$name), ])
8   st_sf(geometry = st_sfc(st_linestring(m[, 1:2]), crs = st_crs
    (place_sf)))
9 }
10
11 path_lusu <- assemble_lusu_routes() # shapefile or waypoint
    polyline
12 path_zhou <- linestring_from_places(loc, c("Poyang_lake", "
    Chaisang", "RedCliffs"))
13 path_zhuge <- linestring_from_places(loc, c("Xiakou", "Chaisang
    ", "RedCliffs", "Fankou"))
14
15 plot(st_geometry(loc), pch = 21, add = FALSE)
16 plot(st_geometry(path_lusu), add = TRUE, col = "#e67e22", lwd
    = 2.8)
17 plot(st_geometry(path_zhou), add = TRUE, col = "#1f77b4", lwd
    = 2.8)
18 plot(st_geometry(path_zhuge), add = TRUE, col = "#2ca02c", lwd
    = 2.8)

```

Listing 3: Map 2: route reconstruction (sf)

```

1 # PSEUDOCODE: see scripts/map3_timeline_routes.R
2 library(sf)
3 library(ggplot2)
4 library(ggrepel)
5

```

```

6 loc <- st_read("Location.geojson", quiet = TRUE)
7 timeline <- data.frame(...) # cols: track, chapter, place,
  stop_idx
8 # join place -> lon/lat; make_segments() -> one row per chapter
  leg
9
10 ggplot() +
11   geom_sf(data = loc) +
12   geom_curve(data = segments_df,
13             aes(x, y, xend, yend, colour = track),
14             arrow = arrow(type = "closed"), linewidth = 1.4) +
15   geom_label_repel(data = chapter_labels, aes(label =
16           chapter_label)) +
17   coord_sf(xlim = c(bbox$xmin, bbox$xmax),
18           ylim = c(bbox$ymin, bbox$ymax), expand = FALSE)

```

Listing 4: Map 3: timeline route map (sf / ggplot2); abbreviated pipeline.

All three spatial outputs in §3.3 use WGS84 (EPSG:4326). Map 1 aggregates interaction events to weighted points (`st_as_sf`); Map 2 connects ordered stops with `st_linestring`(); Map 3 draws chapter transitions with `geom_curve`(). Full scripts are in the project repository.

4. Results

4.1. Network Findings

4.1.1. Centrality Divergence: Degree vs. Eigenvector

Table 1 presents the top-ten characters ranked by weighted degree centrality; eigenvector scores are shown for comparison (highest score held by Lu Su).

Table 1: Top-ten character centrality scores, *Romance of the Three Kingdoms*, Chapters 43 through 50.

Rank	Character	Faction	Degree	Eigenvector
1	Zhuge Liang	Shu	66.8	0.977
2	Zhou Yu	Wu	63.3	0.965
3	Lu Su	Wu	43.8	1.000
4	Cao Cao	Wei	32.9	0.186
5	Sun Quan	Wu	26.9	0.537
6	Liu Bei	Shu	12.2	0.185
7	Huang Gai	Wu	11.9	0.151
8	Zhuge Jin	Wu	10.4	0.210
9	Zhang Zhao	Wu	9.3	0.203
10	Guan Yu	Shu	7.4	0.102

Zhuge Liang and Zhou Yu rank first and second in degree centrality, which reflects their strong narrative presence across Chapters 43–50. However, Lu Su has the highest eigenvector centrality score at a perfect 1.000, showing that he is connected to other highly central characters rather than just having many direct ties. In particular, he is linked to Sun Quan, Zhou Yu, Liu Bei, and Zhuge Liang, the four most important nodes in the coalition network. These connections are mainly diplomatic in nature, which positions Lu Su as the key structural broker in the alliance. (Burt 2004, Chapter 2).

4.1.2. *Sentiment Topology of the Alliance*

The sentiment score of the Zhuge Liang and Zhou Yu edge is +0.11, which suggests a cautious and only weakly cooperative relationship rather than a strongly antagonistic one. Interactions between the two figures are therefore better understood as strategically restrained than openly hostile.

By contrast, Lu Su’s edges present a consistently positive sentiment profile:

- Lu Su and Sun Quan: +1.00 (strongly cooperative)
- Lu Su and Zhuge Liang: +0.71 (cooperative)
- Lu Su and Zhou Yu: +0.64 (cooperative)
- Lu Su and Liu Bei: +0.50 (moderately cooperative)

Lu Su is thus the character whose ties remain most consistently positive across the alliance’s principal actors. Rather than serving as an emotional counterweight, he functions as a diplomatic intermediary whose relationships help maintain communication and alignment across otherwise uneven factional ties.

4.1.3. *Edge Weight Distribution*

A further structural asymmetry emerges when edge weights are disaggregated by interaction type. Lu Su's ties are marked by a strong concentration of Diplomatic and Strategic interactions, especially in his connections with Sun Quan, Zhou Yu, Liu Bei, and Zhuge Liang. By contrast, Zhuge Liang and Zhou Yu are more often involved in interactions coded as Persuasion, Command, Conflict, or Deception, which are important narratively but less consistently oriented toward coalition maintenance. This distribution suggests that Lu Su's centrality is supported by ties that are especially useful for coalition maintenance, rather than by sheer interaction frequency alone.

4.2. Spatial Findings

4.2.1. *Territorial Anchoring vs. Diplomatic Mobility*

The route reconstruction maps (Figure 3 and Figure 4) reveal three distinct spatial logics operating within the same alliance:

Zhou Yu: Anchored Commander. Zhou Yu's movement is concentrated in Wu coastal and river-mouth territories: Chaisang, Poyang Lake and the Red Cliffs naval staging zone. His spatial footprint is deep within Wu territory. He is a territorial commander whose authority is spatially bounded and whose strategic legitimacy derives from control of the naval theatre.

Zhuge Liang: Long-Range but Episodic. Zhuge Liang covers the greatest single-journey distances, notably the embassy crossing from Liu Bei's camp at Xiakou to Sun Quan's court at Chaisang, but his spatial presence within Wu is *episodic*: he arrives, performs his diplomatic mission and withdraws. He lacks territorial anchoring within the Wu spatial domain, which limits his structural embeddedness in the coalition.

Lu Su: Mobile Diplomatic Infrastructure. Lu Su's trajectory is the most spatially continuous. He travels between Liu Bei's camp, Sun Quan's court, Zhou Yu's military headquarters and multiple deliberation sites, maintaining connection across all spatial sub-zones of the coalition. His route-distance across coalition-critical spatial transitions exceeds that of both Zhuge Liang and Zhou Yu, and crucially his traversals occur more evenly across the temporal span of the eight chapters, suggesting sustained rather than punctual diplomatic labour.

4.2.2. *Chaisang as Spatial Centroid*

Kernel density estimation of person to place interaction events identifies Chaisang as the spatial centroid of coalition formation activity: it is the location with the highest

aggregate interaction weight (score = 4.7 on a relative 0 to 5 scale), serving as site for Sun Quan’s deliberations, Lu Su’s briefings, Zhou Yu’s military planning and Zhuge Liang’s persuasive oratory. Its spatial position (at the confluence of Wu territory and the Yangtze access corridor to Red Cliffs) makes it a logistical hub as well as a political one.

4.2.3. *The Yangtze River as Narrative Boundary*

The Yangtze River functions in the spatial analysis as more than a geographic feature. In the interaction heat map, it serves as a visible boundary separating Wu’s high-density activity cluster from Cao Cao’s isolated northern encampment at Wulin. The river marks a narrative logic of *coalition cohesion* on its southern bank versus *strategic isolation* on its northern shore: a spatial encoding of the very political asymmetry that made the fire-attack strategy possible.

Red Cliffs itself, despite being the episode’s climactic location, is not a high-density interaction site in the person to place network: it is the endpoint of converging trajectories, a zone of execution rather than negotiation. The coalition is *formed* in Wu’s interior (Chaisang); it is only *deployed* at Red Cliffs. This spatial distinction challenges readings that treat the battle site as the alliance’s constitutive moment.

5. Discussion

5.1. Lu Su as Structural Broker

Our network results support a coherent reading of Lu Su as a diplomatic broker in the sense of structural hole theory (Burt 2004). A structural hole exists when two otherwise weakly connected nodes depend on a third node to relay information, resources, or trust. In this case, the Zhuge Liang and Zhou Yu relationship is better understood as a constrained cross-factional tie rather than a fully stable alliance link. Lu Su occupies the bridging position between them and the wider coalition not simply by having many contacts, but by maintaining ties to the key figures on both sides. His perfect eigenvector centrality reflects this position: he is connected to the principal agents of both allied factions through ties that are sufficiently cooperative to sustain coordination. In graph-theoretic terms, Lu Su is the node whose removal would most weaken the coalition’s connective structure. This reading aligns with the broader historical sense that Lu Su was indispensable to alliance maintenance, even if the novel’s narrative emphasis tends to foreground Zhuge Liang and Zhou Yu instead.

5.2. Fragile Alliance Theory

Our integrated network-spatial analysis supports what may be called a *fragile convergence* model of the Sun Liu alliance. The coalition is not best understood as a seamless merger of strategic will, but as a temporary alignment of separate political and spatial logics, held together by forms of mediation that remain unstable and contingent.

Three structural features contribute to this fragility:

1. **Sentiment unevenness at the core dyad.** The Zhuge Liang and Zhou Yu relationship is not uniformly cooperative, but marked by a cautious and only weakly positive pattern of interaction. Alliance cohesion therefore does not rest on a single harmonious core, but on Lu Su's ability to maintain workable ties across the principal figures of the coalition.
2. **Spatial non-integration.** The coalition never achieves full spatial interpenetration. Wu and Shu characters occupy distinct territorial zones; their convergence at Red Cliffs is momentary, not structural. Once the battle concludes, the coalition geography immediately reasserts itself, as subsequent chapters confirm.
3. **Asymmetric dependence.** Liu Bei and Zhuge Liang need Wu far more than Wu needs them. This asymmetry is encoded in the spatial mobility data: Shu figures travel into Wu territory; Wu figures do not travel into Shu. Lu Su's role is partly to manage the emotional and diplomatic consequences of this imbalance, which his consistently high cooperative sentiment scores reflect.

This fragile-convergence reading has implications beyond the narrative. The historical Battle of Red Cliffs was indeed followed by years of territorial contention between Sun Quan and Liu Bei, during which Lu Su famously worked to maintain the alliance against his own faction's preference for reclaiming Jing Province. Our computational reading of the novel's narrative structure thus resonates with the political history it fictionalises.

5.3. Integrating Network and Spatial Evidence

The dual-method design yields interpretive dividends that neither approach alone could produce. Network analysis establishes *who* is structurally indispensable; spatial analysis explains *why*: Lu Su's eigenvector centrality is sustained by his continuous traversal of coalition geography, which allows him to carry political intelligence, emotional reassurance and diplomatic legitimacy across the spatial boundaries that separate his principals.

Zhugge Liang's long-range mobility could in principle achieve similar effects, but his episodic presence within Wu territory means he cannot sustain the ongoing relationship maintenance that coalition stability requires. Zhou Yu's deep territorial embeddedness gives him military authority but precludes the inter-factional mobility that brokerage

demands. Only Lu Su moves continuously enough and widely enough to serve as the coalition’s diplomatic circulatory system.

5.4. Narrative vs. Structural Visibility

A meta-finding of this study is methodological: narrative prominence and structural centrality diverge in systematic ways that DH methods can surface. The *Romance of the Three Kingdoms* is constructed around a rhetoric of strategic genius and martial virtue that centres Zhuge Liang and Zhou Yu as its dramatic engines. This rhetoric serves the novel’s affective purposes: readers experience the alliance through these figures’ competing brilliances, but it systematically displaces the unglamorous work of trust-building, message-carrying and relationship-maintenance that Lu Su performs.

Moretti’s argument that distant reading shifts attention from close analysis of individual texts to patterns visible at the scale of the literary system (Moretti 2005) finds concrete illustration here. A close reader of Chapters 43 through 50 might intuit Lu Su’s importance; the network analysis makes it precise, reproducible and theoretically grounded.

6. Limitations

6.1. Narrative Bias in Sentiment Coding

Sentiment scores are derived from textual events as rendered in English translation. Moss Roberts’s translation, while scholarly, makes interpretive choices that may inflate or deflate the hostility of particular interactions. Moreover, the Cooperative / Neutral / Hostile trichotomy is a coarse instrument: an interaction can be formally cooperative (an agreement is reached) while containing undercurrents of mutual mistrust that the coding scheme cannot capture. Future work should explore finer-grained sentiment lexicons calibrated to Classical Chinese rhetorical conventions.

6.2. Uncertainty in Coordinate Mapping

Assigning geographic coordinates to narrative locations involves acknowledged imprecision. While major locations (Chaisang, Red Cliffs, Xiakou) have well-attested approximate positions, smaller sites named in the chapters may correspond to multiple candidates in the historical geography, or may be entirely fictional. We recorded coordinate confidence levels, but the spatial analysis should be understood as a model of relative spatial logic rather than an accurate historical-geographic reconstruction.

6.3. LLM Extraction Limitations

Large-language-model extraction of character interactions, despite multi-model validation and manual correction, introduces two systematic biases. First, LLMs tend to over-extract interactions involving characters who receive heavy narratorial attention, which may inflate the degree centrality of already prominent characters. Second, indirect interactions (where a character acts on another’s behalf, or is the topic of discussion between third parties) are inconsistently captured. These limitations particularly affect the measurement of Lu Su’s influence, which is partly exerted through behind-the-scenes diplomatic preparation that the novel narrates obliquely rather than directly. Our manual correction stage addressed the most egregious cases but cannot eliminate the pattern entirely.

6.4. Single-Text Corpus

Our analysis is based on one translation of one literary work. Cross-linguistic validation (using the original classical Chinese text) and cross-textual validation (against the historical *Records of the Three Kingdoms*) would strengthen the generalisability of the findings. Zhang et al. (2021) provide a framework for such comparison that future work could extend to the specific episode analysed here.

7. Conclusion

This paper has applied a hybrid Digital Humanities methodology, combining LLM-assisted interaction extraction, network centrality analysis, and GIS-based spatial mapping, applied to Chapters 43 through 50 of *Romance of the Three Kingdoms*. Our principal finding is that Lu Su, not Zhuge Liang or Zhou Yu, functions as the structural broker of the Sun Liu alliance. His maximum eigenvector centrality score, his consistently cooperative sentiment profile across all principal dyads and his unmatched spatial continuity across coalition geography converge on a single interpretive conclusion: the alliance at Red Cliffs is structurally dependent on one character whose labour the novel systematically undervalues.

This *fragile convergence* reading carries implications for how we understand pre-modern Chinese alliance politics as novelistically encoded. The *Romance*’s rhetorical economy: its investment in martial brilliance, strategic genius and the charismatic duel of Zhuge Liang and Zhou Yu, renders invisible the unglamorous but essential work of diplomatic maintenance. Computational analysis makes that invisible labour visible, precise, and theoretically articulable.

More broadly, this study demonstrates that Digital Humanities methods bring genuine interpretive returns to classical Chinese fiction, a corpus that has received less compu-

tational attention than European canonical texts. The hybrid pipeline developed here (controlled interaction extraction, weighted sentiment-coded networks, and narrative-space GIS projection) is readily adaptable to other episodes, other novels and other pre-modern narrative traditions. Future work might extend the spatial analysis to the full *Romance* to examine how alliance geographies evolve across its 120 chapters, or apply the brokerage-identification framework to other coalition episodes in Chinese historical fiction.

Data and Code Availability. Interaction datasets, R analysis scripts, and QGIS layers are available at https://github.com/nhatminh37/HUMA_1678.

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